

The acute effects of community violence on young children's regulatory, behavioral, and developmental outcomes in a low-income urban sample in Brazil

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Background: Existing research on the impacts of adversity on young children's psychological well-being has largely focused on household-level risk factors using observational methods in high-income countries. This study leverages natural variation in the timing and location of community homicides to estimate their acute effects on the regulatory, behavioral, and developmental outcomes of Brazilian 3-year-olds. **Methods:** We compared the outcomes of children who were assessed soon after a recent neighborhood homicide to those of children from the same residential neighborhoods who had not recently experienced community violence. Our sample included 3,241 3-year-olds ($M_{\text{age}} = 41.05$ months; 53% female; 45% caregiver education less than middle school; 26% receiving a public assistance program) from seven neighborhoods in São Paulo, Brazil. Child outcome measures included parent reports of effortful control and behavior problems as well as direct assessments of children's developmental (cognitive, language, and motor) skills. Community homicides were measured using police records. **Results:** Recent exposure to community homicides was associated with lower effortful control, higher behavior problems, and lower overall developmental performance for children ($d = .05$ –.20 standard deviations; $p = ns - <.001$). Effects were consistent across subgroups based on sociodemographic characteristics and environmental supports, but generally largest when community violence exposure was geographically proximal (within 600 m of home) and recent (within 2 weeks prior to assessment). **Conclusions:** Results highlight the pervasive effects that community violence can have on young children as well as the need to expand support to mitigate these effects and prevent inequities early in life. **Keywords:** Violence; early life experience; adversity.

Introduction

The first years of life are a sensitive period during which children's environmental experiences can profoundly impact their long-term outcomes. Exposure to adversity in this period has been linked to reductions in children's early regulatory, behavioral, and cognitive skills, with implications for later-life psychological well-being (Nelson, 2013; Shonkoff, Boyce, & McEwen, 2009). Importantly, much of the psychological research on early childhood adversity has focused on household-level risks (e.g., household poverty, corporal punishment), with relatively little consideration of how broader ecological threats may shape children's outcomes (McLaughlin, Weissman, & Bitrán, 2019).

Community violence—or neighborhood-based acts of interpersonal violence such as homicide or assault—is a prevalent and impactful form of adversity for many children. Children's experiences of community violence can range from direct victimization to more indirect forms of exposure, including witnessing or hearing about a crime or simply observing the social-emotional “aftershocks” that a violent event might have on community residents (e.g., increases in

caregiver anxiety, heightened police presence; McCoy, 2013). In turn, children's behavioral well-being has been found to change as a result of community violence exposure, both chronic (e.g., Chen, Voisin, & Jacobson, 2016; Mohammad, Shapiro, Wainwright, & Carter, 2015) and acute (e.g., Cuartas & Leventhal, 2020; Molano, Harker, & Cristancho, 2018; Sharkey, 2010). For example, a landmark study by Sharkey (2010) using data from Chicago, US, showed short-term negative effects of a recent residential neighborhood homicide—an especially salient and threatening form of community violence—on adolescents' academic performance. Research from Colombia has also shown the acute impacts of police-reported residential community violence on adolescents' reports of overall mental health as well as specific measures of emotion regulation, empathy, and avoidant behavior (Cuartas & Leventhal, 2020; Molano et al., 2018).

Although the mechanisms underlying these acute effects remain poorly understood, bioecological theories from multiple fields (e.g., criminology, developmental psychology, neuroscience) suggest that increases in perceived stress and neuroendocrine activity following violence exposure could disrupt test performance and behavior via both neurobiological and environmental pathways (Defoe, 2021;

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McCoy, 2013; McLaughlin & Sheridan, 2016). Neurobiologically, community violence has been shown to acutely impair children's sustained and selective attention and impulse control (McCoy, Raver, & Sharkey, 2015; Sharkey, Tirado-Strayer, Papachristos, & Raver, 2012), which in turn could compromise their ability to appropriately regulate their emotions and behaviors. Recent community violence exposure has also been linked to short-term increases in parents' depressive symptoms and reductions in their provision of stimulating activities (Cuartas, McCoy, & Molano, 2018; Sharkey et al., 2012), environmental pathways with both immediate and longer-term implications for children's well-being, especially during the early childhood period (Britto et al., 2017).

Importantly, most research on community violence has focused on adolescent outcomes, with relatively little consideration of its impacts during the first years of life (Leventhal, 2018). Although young children are less likely than their older peers to directly witness or be victimized by a violent crime (Finkelhor, Turner, Shattuck, & Hamby, 2015), even indirect forms of violence could have meaningful consequences during this sensitive period of development in which children are deeply attuned to environmental instability (McCoy, 2013). Furthermore, despite the fact that both exposure to and consequences of violence exposure are likely to vary cross-culturally (Defoe, 2021), research on community violence has focused on a small number of settings. Apart from several studies in Colombia (e.g., Cuartas et al., 2018; Molano et al., 2018; Obando, Wright, & Hill, 2023), there is little evidence regarding the impacts of violence in low- and middle-income countries, where supports for families are often limited and levels of violence are especially high. Globally, most of the more than 1 billion children ages 2–17 who have been exposed to violence live in a low- or middle-income country (Hillis, Mercy, Amobi, & Kress, 2016). The homicide rate in Brazil, for example, is more than five times the global average (27 vs. 5 per 100,000 people; UN Office on Drugs & Crime, 2021), suggesting that the collective consequences of violence may be particularly profound in this context.

Community violence research has also faced key methodological challenges that limit the credibility of many existing inferences. Like most forms of adversity, community violence does not occur at random. As such, it is difficult to empirically disentangle the effects of violence from those of co-occurring risk factors (e.g., household poverty, low community resources, etc.; Leventhal, Dupéré, & Shuey, 2015). Furthermore, much of the existing literature has relied on participant (e.g., parent, child) reports of violence exposure to understand the implications for child outcomes (Leventhal et al., 2015). This approach has been criticized for both its lack of reliability and its likely bias due to measurement

error, whereby individuals who experience negative psychological consequences of violence may systematically over- or underreport their exposure (McCoy, 2013).

This study aims to address these gaps in the literature by examining the acute impacts of exposure to recent community violence (residential neighborhood homicide) on low-income 3-year-olds' regulatory, behavioral, and developmental outcomes in seven socioeconomically diverse communities in São Paulo, Brazil. In addition to focusing on an underrepresented age period (early childhood) and context (Brazil), this study leverages natural, arguably random variation in the timing of homicides relative to children's assessments to estimate the impacts of recent exposure to community violence on children's immediate outcomes. Furthermore, by using police- rather than child- or parent reports of crime, we aim to more objectively and comprehensively capture the consequences of both direct and indirect exposure to community violence, including the potential effects for children who were not consciously aware that a homicide had taken place. We compare the impacts of community violence based on (a) how long ago the homicide occurred, (b) how far away the homicide was from the child's home, and (c) the specific outcome under consideration. Finally, we explore differential effects across subgroups of children based on their sociodemographic backgrounds and exposure to other environmental supports and risks. We hypothesize statistically detectable differences in children's regulatory, behavioral, and developmental outcomes based on violence exposure, with the strongest impacts occurring for children whose exposure was geographically proximal and temporally recent.

Methods

Sample and procedures

Child-level data were taken from an ongoing longitudinal study of early life adversity and child outcomes called the São Paulo Western Region Birth Cohort ("ROC study"), which enrolled 6,162 low-income children born at the publicly funded University Hospital of the University of São Paulo (HUUSP) between April 1, 2012 and March 31, 2014. Data for the present study come from the 3-year follow-up of the ROC study and include information on children's outcomes and socio-demographic characteristics collected by both parent reports and direct assessments led by trained enumerators in children's homes. Data also included children's residential addresses. A detailed description of the ROC study can be found in Brentani et al. (2020).

The analytic sample for the present study includes 3,241 children from the 3-year follow-up (53% of the original sample). The remaining 2,921 children were either lost to follow-up due to administrative challenges (e.g., local strikes, staff turnover, difficulties locating families; Brentani et al., 2020; $n = 2,543$; 87%) or lacked a residential address that could be accurately geocoded and had at least one available outcome measure at the 3-year follow-up ($n = 378$; 13%). Children included in our analytic sample were generally similar to those who left the

original sample in terms of baseline sociodemographic characteristics (see Table S1 for details).

Demographic characteristics of the analytic sample are shown in Table 1. Children were on average 41.05 months old ($SD=7.35$) at the time of the 3-year assessment, and 53% were female. Although racial/ethnic categories are known to be unreliable in Brazil (Cardena et al., 2013) and therefore not individually reported in this study, self-reports from a selected subgroup of participants suggest that the sample was racially and ethnically diverse. Children's primary caregivers were on average 33.01 years old ($SD=11.53$). Reflecting their recruitment from a publicly funded hospital in a socioeconomically diverse part of São Paulo, 4% of caregivers had completed no education (compared with 33% of women ages 30–34 nationally and 24% in the state of São Paulo), 41 percent had completed elementary school (vs. 17% both nationally and statewide), 50% had completed middle school (vs. 35% nationally and 38% statewide), and 5% had completed upper/tertiary education (vs. 16% nationally and 20% statewide; IBGE, 2010). Caregivers worked an average of 17.89 hr/week ($SD=20.59$). Fathers were present in 68% of households, and 26% of households were enrolled in the Bolsa Familia public cash transfer assistance program for low-income families. On average, caregivers reported using relatively few forms of physical punishment on children at home ($M=0.84$, $SD=1.01$, of six items reported).

Child-level data were merged with community information from two public sources. First, the dates and addresses of police-reported homicides in the municipality of São Paulo were taken from the State Secretariat of Public Safety (www.ssp.sp.gov.br). Second, district boundaries within the city of São Paulo were taken from the 2010 population census collected by the Brazilian Institute of Geography and Statistics (ibge.gov.br).

Measures

Child outcomes. Three child outcome measures were used from the 3-year follow-up of the ROC study. First, children's self-regulation was measured using the effortful control subscale of the Early Childhood Behavior Questionnaire (ECBQ; Putnam, Gartstein, & Rothbart, 2006), a 13-item parent report of children's ability to regulate behavioral and emotional responses during the preceding 6 months ($\alpha=.74$). Second, children's behavior problems were measured using the Strengths and Difficulties Questionnaire (SDQ; Goodman, 1997), a parent report of internalizing and externalizing behavior in the prior 6 months ($\alpha_{\text{overall}}=.74$; $\alpha_{\text{internalizing}}=.59$; $\alpha_{\text{externalizing}}=.67$). Finally, we measured children's development using the Engle scale of the Regional Project on Child Development Indicators (PRIDI for its Spanish acronym; Verdisco et al., 2009), a direct assessment of children's holistic cognitive, language, and motor skills ($\alpha_{\text{overall}}=.93$; $\alpha_{\text{cognitive}}=.86$; $\alpha_{\text{language}}=.88$; $\alpha_{\text{motor}}=.76$).

Details of all child outcome measures, including summary statistics and psychometric properties, are provided in Appendix S1 and Tables S2–S4.

Community violence exposure. We used publicly available police reports of community violence from the Secretariat of Public Safety of the State of São Paulo. Consistent with prior research (e.g., Molano et al., 2018; Sharkey, 2010), we focused on homicides—including willful murder (92.9%), robbery followed by death (6.2%), bodily injury followed by death (0.8%), and death resulting from police intervention (0.1%)—to represent community violence, as we hypothesized these crimes to be the most reliably reported and most salient, threatening, and stressful to community residents.

To quantify children's exposure to community violence, we began by geocoding the addresses of children's homes and the location of homicides using the R *ggmap* package (Kahle & Wickham, 2013). We then calculated the number of homicides that occurred within five geographic radii around each child's home (600, 700, 800, 900 m, and 1 km) and three temporal windows leading up to the child's 3-year assessment (2, 3, and 4 weeks) using QGIS software (QGIS Development Team, 2021). We selected these geographic radii and temporal windows to balance likely impacts of community violence with sample size considerations. Specifically, although the literature suggests that the acute effects of community violence on child outcomes may be stronger for crimes that occur in close geographic and temporal proximity (Sharkey, 2010), we did not include geographic radii less than 600 m or temporal windows of less than 2 weeks because fewer than 5% of children in our sample experienced any crimes within these radii/windows (see Tables S5–S8).

Using this information, we then generated a total of 15 indicators of exposure for each child (five geographic radii times three temporal windows), where a child was “exposed” to community violence if *at least* one homicide occurred within the given geographic radius in the specified time period leading up to the child's assessment. We chose to express exposure in a binary fashion rather than as the total number of crimes because the latter had very little variability (i.e., very few exposed children experienced more than one homicide). Children who had no violent crimes occur within the given geographic radius/temporal window were considered “unexposed.” Consistent with prior work (Cuartas et al., 2018; McCoy et al., 2015), to avoid spillover effects (i.e., contamination), we excluded from the control group children who were exposed to a crime in the week before the specified temporal window of interest (e.g., children who experienced a crime between 2 and 3 weeks prior to their assessment for the 2-week window) with the expectation that these children may still be experiencing some lingering effects of the crime. We also excluded children who were assessed on the same day as the crime, as we did not know whether exposure took place before or after the assessment.

Figure 1 illustrates our assignment strategy of children to the “exposed” and “unexposed” groups based on geography.

District boundaries. District boundaries were taken from Brazil's 2010 population census. Of the 96 districts in the city of São Paulo, children in our study lived in seven. These seven districts had an average population of 86,442 residents ($SD=34,834$) and an average size of 1,043 hectares ($SD=206$).

Covariates. Our analyses included child (age, sex, number of days per week attending childcare or “crèche”) and caregiver characteristics (educational attainment) as covariates. Consistent with other research from low- and middle-income countries (e.g., Filmer & Pritchett, 2001; Hamadani et al., 2010; Kariger et al., 2012), we also controlled for household socioeconomic status (SES) based on the first principal component of various household assets and an overcrowding index, as well as children's home learning environment based on an index of household play materials (the total from four different types of toys: homemade, manufactured, household objects, and objects found in the street). All these covariates were reported by the child's primary caregiver at the 3-year follow-up. We also included a “broken home index” designed to measure housing quality, which was composed of 10 binary items observed by data collectors during the home visit (e.g., “Were there any broken windows?”). Finally, we included a measure of chronic exposure to community violence (constructed as the total number of homicides that occurred in the given geographic radius since the child's birth) to control for any outcome

Table 1 Sample descriptive statistics and balance across community violence-unexposed and exposed children (600 m and 2 weeks)

Variable	Full Sample			Unexposed			Exposed			Covariate differences based on exposure	
	<i>N</i>	% Missing	Mean	<i>SD</i>	<i>n</i>	Mean	<i>SD</i>	<i>n</i>	Mean	<i>SD</i>	<i>p</i> -Value <i>d</i>
Child characteristics											
Chronic crime exposure (number of homicides in 600 m radius surrounding home since child's birth) ^a	3,122	0.00	7.68	6.30	2,966	7.43	6.07	156	12.33	8.46	.000
Age at which child was assessed (month) ^a	3,122	0.00	41.05	7.35	2,966	40.99	7.26	156	42.10	8.84	.328
Child is female ^a	3,122	0.00	0.53	0.50	2,966	0.53	0.50	156	0.55	0.50	.670
Number of days in crèche (week) ^a	2,818	9.74	4.18	1.86	2,675	4.19	1.85	143	3.99	2.03	.395
Caregiver characteristics											
Caregiver's age (years)	2,254	27.80	33.01	11.53	2,138	33.04	11.59	116	32.47	10.42	.606
Caregiver has no education ^a	2,999	3.94	0.04	0.18	2,852	0.04	0.19	147	0.01	0.08	.000
Caregiver's highest grade completed = elementary ^a	2,999	3.94	0.41	0.49	2,852	0.41	0.49	147	0.41	0.49	.940
Caregiver's highest grade completed = middle ^a	2,999	3.94	0.50	0.50	2,852	0.50	0.50	147	0.51	0.50	.623
Caregiver's highest grade completed = upper ^a	2,999	3.94	0.06	0.23	2,852	0.05	0.23	147	0.07	0.25	.532
Number of hours/week caregiver works	2,804	10.19	17.89	20.59	2,667	17.77	20.53	137	20.20	21.74	.352
Household characteristics											
Father lives in household	3,108	0.45	0.68	0.46	2,952	0.69	0.47	156	0.68	0.47	.853
Father has been to prison	3,064	1.86	0.10	0.30	2,913	0.10	0.30	151	0.11	0.31	.824
SES index (z-score) ^a	2,808	10.06	0.01	1.00	2,666	0.01	1.01	142	-0.03	0.94	.824
Household receives support from Bolsa Familia	3,085	1.19	0.26	0.44	2,930	0.26	0.44	155	0.26	0.44	.823
Number of picture books	2,926	6.28	7.28	32.43	2,782	7.29	33.20	144	7.13	8.25	.005
Toy index (total out of 4) ^a	3,090	1.02	2.42	0.99	2,936	2.42	1.00	154	2.52	0.93	.541
Physical punishment index (total out of 6)	3,096	0.83	0.84	1.01	2,941	0.84	1.00	155	0.92	1.01	.343
Broken home index (average out of 10) ^a	3,039	2.66	0.13	0.23	2,887	0.14	0.23	152	0.10	0.19	.030
Child outcomes											
Effortful control (average)	3,080	1.35	4.98	1.10	2,925	4.99	1.10	155	4.87	1.15	
Behavior problems (average)											
Total problems	3,090	1.02	0.47	0.31	2,936	0.46	0.30	154	0.53	0.33	
Internalizing problems	3,092	0.96	0.32	0.30	2,938	0.31	0.29	154	0.36	0.33	
Externalizing problems	3,093	0.93	0.62	0.41	2,939	0.61	0.41	154	0.69	0.42	
Development (average)											
All domains	2,390	23.45	0.67	0.18	2,270	0.67	0.18	120	0.64	0.19	
Cognitive	2,574	17.55	0.67	0.22	2,444	0.67	0.22	130	0.64	0.24	
Language	2,684	14.03	0.68	0.21	2,548	0.68	0.21	136	0.65	0.22	
Motor	2,656	14.93	0.63	0.19	2,526	0.63	0.19	130	0.61	0.19	

^aVariable was included as covariate in main regression analyses (Figure 2). In the regression analyses, caregiver education was expressed as an ordinal variable from 0 (no education) to 3 (completed upper/tertiary education); the SES index was constructed by principal component analysis and includes various household assets and an overcrowding indicator; Toys index reflects the total number of different household play materials/toys available in the home (homemade, manufactured, household objects, and objects found in the street); physical discipline index includes a caregiver-reported count of the number of forms of discipline (e.g., spanking) used on the child at home (out of 6) by the caregiver or another adult in the house in the last 30 days; Broken home index reflects housing quality and is composed of 10 binary items observed by data collectors during the home visit.

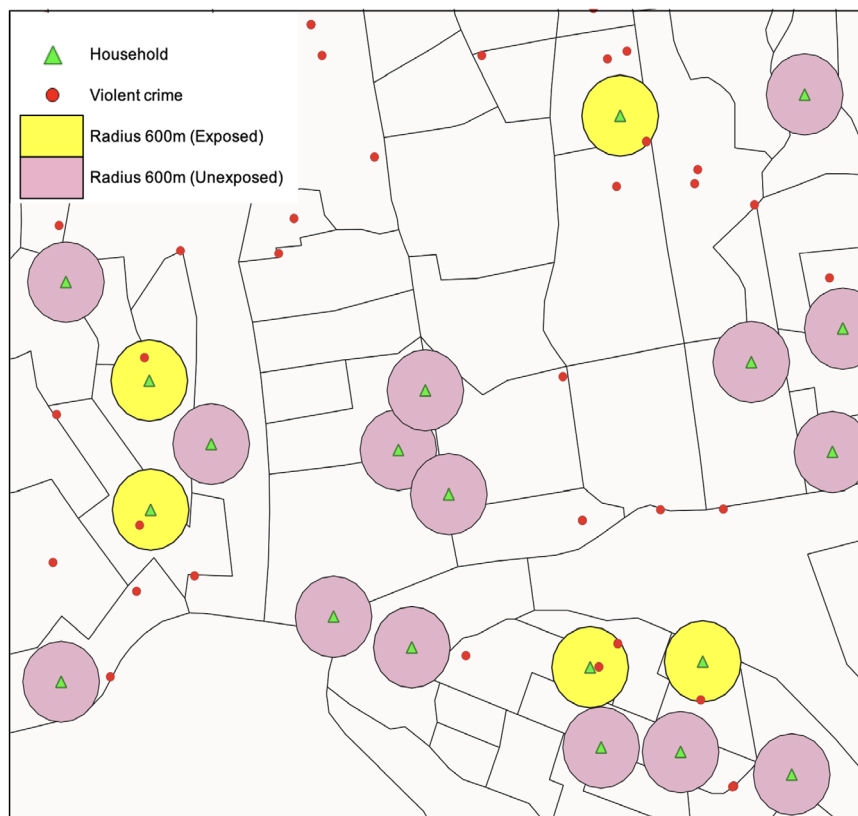


Figure 1 Illustration of children's classification as exposed vs. unexposed to community violence based on the geographic distance of their household from a violent crime. *Note.* Figure is for illustrative purposes only. Map represents an area of São Paulo. Locations of children's homes are approximate

differences attributable to ongoing exposure to violence rather than a recent, acute shock.

Analytic approach

We first checked the balance of observable characteristics between children assigned to the exposed and unexposed groups. We then estimated the effects of community violent crime on children's outcomes using the following linear regression equation:

$$Y_{ij} = \alpha_d + \beta_1 \text{Exposed}_{ij} + \beta_2 \mathbf{X}_{ij} + \epsilon_{ij}, \quad (1)$$

where Y is the outcome of child i in district j as measured by the ECBQ effortful control subscale score (self-regulation), the SDQ total behavior problems score (behavior), or the PRIDI total score (development; all expressed as z-scores); Exposed_{ij} is a binary variable equal to 1 if the child was exposed to a violent crime in the given geographic radius/temporal window of interest; and $\mathbf{X}_{ij}$ is a vector of child and household covariates described above. The latter were chosen based on evidence for their imbalance between exposed and unexposed children (see Table 1 and Tables S9–S22) and/or evidence from literature in low- and middle-income countries suggesting that they are potential confounders in our model (e.g., Fernald, Kariger, Hidrobo, & Gertler, 2012; Frongillo, Kulkarni, Basnet, & de Castro, 2017; McCoy et al., 2016). The parameter of interest was β_1 , which estimates the relationship between the given definition of community violence exposure and the child outcome. All regressions included district fixed effects, α_d , to control for broader time-invariant geographical characteristics that may affect child outcomes (e.g., community resource availability) and robust standard errors.

We also explored possible variability in the effect of community violence exposure by supplementing equation (1) with an interaction between Exposed_{ij} and the following subgrouping variables: child sex (1 = female), caregiver education (1 = completed middle school or less), household SES (1 ≤ 50th percentile), receipt of the Bolsa Familia social assistance program (1 = family receives), social support (1 ≤ 50th percentile on an index of social support constructed as the total out of four types of personal supports available to the caregiver, e.g., someone to loan a small amount of money if needed), and chronic community violence exposure (1 ≤ 50th percentile on the sum of homicides in the geographic radius of interest since birth).

Missing data. We performed multiple imputations by chained equations (MICE) to handle missing data on both our outcomes and covariates. Table 1 includes the sample size for each of our key variables, with missingness rates ranging from 1% to 10% across variables included as covariates in our regression analyses. Multiple imputation was used to address missingness given its efficiency, ability to use all available data, and preservation of sample size (Lall, 2016; McCleary, 2002).

Sensitivity analyses. We conducted sensitivity analyses to test the robustness of our primary findings to alternative modeling specifications. Specifically, we examined results when (a) focusing on the various subscales of the SDQ and PRIDI (rather than their total scores), (b) using listwise deletion for outcome variables and imputation for covariates only (rather than imputing both outcomes and covariates), and (c) limiting the unexposed group to children who had experienced a violent crime in the 2–4 weeks following the assessment (rather than including all children who had simply not

experienced exposure during the focal temporal window). Although limited in statistical power and external validity, this final sensitivity analysis allowed us to create an unexposed (control) group that may be more similar to the exposed (treatment) group in ways other than the *timing* of their exposure.

All analyses were conducted in Stata 17 (StataCorp, 2021).

Results

Balance in observables

Results of analyses examining balance of observable characteristics across all definitions of community violence exposure are shown in Tables S9–S22, with Table 1 summarizing balance for exposure within 600 m and 2 weeks. The variable showing the most consistent evidence of imbalance was chronic community violence, with children exposed to a recent homicide experiencing significantly higher average levels of chronic community violence compared with their recently unexposed peers, regardless of the definition of exposure (range $d = .54$ to $.72$, all $p < .001$). Although no other variables showed consistent evidence of imbalance, some did emerge as statistically significantly different across groups for certain exposure definitions. For example, children who were exposed to community violence within 600 m of their homes and 2 weeks of their assessment were less likely than their unexposed peers to have a caregiver with no education and to live in a low-quality house based on the broken home index (see Table 1). These variables—as well as the covariates described above—were included in subsequent regression analyses to improve the precision of estimates and account for potential imbalance.

Estimates of the effects of community violence exposure on child outcomes

Results of our primary regression analyses estimating the impacts of recent community violence exposure on children's outcomes are visualized in Figure 2 and detailed in Tables S23–S25. Overall, children who were exposed to at least one homicide close to their homes in the weeks prior to their 3-year assessment exhibited lower levels of effortful control, higher levels of behavior problems, and lower performance on a standardized developmental assessment relative to their recently unexposed peers from the same communities. These associations were mostly—but not always—statistically significant. Effect sizes were relatively modest, ranging from approximately 0.05 SD ($p = ns$) to 0.20 ($p < .01$), depending on the outcome and definition of exposure. For behavior problems and overall development scores, the effects of community violence were generally stronger when exposure took place nearer to children's homes and closer in time to their assessment. (Effects for effortful control were somewhat more stable across space and time.)

Results of analyses examining subgroup differences are shown in Tables S26–S31. Overall, we found no consistent evidence for systematic variability in the effects of community violence exposure based on children's sociodemographic background or their exposure to other environmental supports and risks. Specifically, just 11 of the 270 interactions examined (15 definitions of exposure times 3 outcomes times 6 subgrouping variables) were statistically significant at $p < .10$, which is below what would be expected by chance, and only one remained significant when adjusting for multiple hypothesis testing using sharped False Discovery Rate q -values (Anderson, 2008).

Sensitivity analyses

Results of sensitivity analyses are detailed in Figures S1–S3 and Tables S32–S34. In general, these results support the conclusions of our primary models. Estimates were robust to the exclusion of cases that were missing child outcome data. Although effects were generally consistent across different subdomains of child development using the PRIDI (i.e., cognitive, language, and motor skills), we found somewhat larger and more robust impacts on children's externalizing behavior problems (range $d = .06$ to $.21$) relative to internalizing behavior problems (range $d = .04$ to $.13$), especially for children whose exposure was geographically and temporally proximal. Finally, analyses limiting the unexposed group to a subset of children who *did* experience violence postassessment showed somewhat stronger effects of recent violence exposure on child development (max $d = -.30$), slightly weaker effects on effortful control (max $d = -.17$), and generally null effects on behavior problems (max $d = .09$).

Discussion

This study leveraged plausibly random variation in the timing of police-reported homicides to estimate the acute impact of recent exposure to community violence on the outcomes of low-income 3-year-olds living in seven socioeconomically diverse communities in São Paulo, Brazil. We found that recent community violence exposure had a negative impact on children's regulatory, behavioral, and developmental outcomes, with effect sizes ranging from approximately 0.05 to 0.20 SD s. Effects on children's development scores, for example, were equivalent to approximately 7 months of learning losses. Additionally, these effects did not systematically vary based on children's sociodemographic characteristics or exposure to other community supports and risk factors. These findings are in line with prior research on older children and adolescents from other countries (e.g., United States, Colombia; McCoy et al., 2015; Molano et al., 2018; Sharkey, 2010) and reinforce the pervasive and

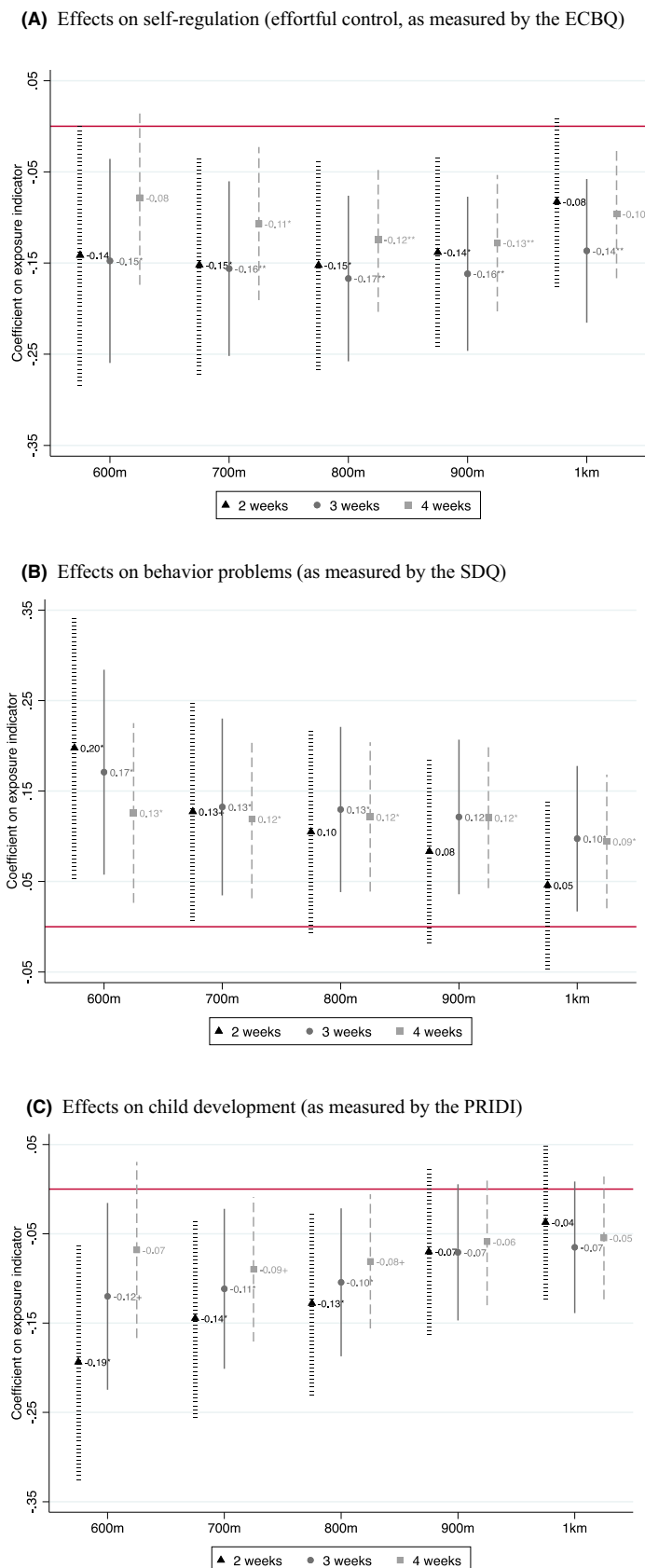


Figure 2 Estimates (with 90% confidence intervals) of the acute effect of exposure to recent community violence on children's outcomes across different geographic radii and temporal windows. (A) Effects on self-regulation (effortful control, as measured by the ECBQ). (B) Effects on behavior problems (as measured by the SDQ). (C) Effects on child development (as measured by the PRIDI). *Note.* Horizontal (red) lines align with a value of zero for the effect of exposure. Vertical lines represent 90% confidence intervals of the exposure coefficients. When the confidence interval overlaps with the horizontal (red) line, the coefficient on the exposure indicator is not statistically significant. + $p < .10$; * $p < .05$; ** $p < .01$

potentially universal consequences that community violence can have for low-income individuals. Importantly, our results are the first to suggest that community violence can have detectable impacts on very young children, who are even less likely than their older peers to be consciously aware that a homicide has taken place.

Like prior work (e.g., Sharkey, 2010), we also found that the impacts of community violence on children's behavioral and developmental outcomes tended to be strongest when exposure was geographically proximal and recent. Negative impacts on effortful control, however, were more consistent across the spatial and temporal windows included in this study. Collectively, these more durable impacts on effortful control are consistent with the neurobiological pathways hypothesized by many bioecological models of violence exposure (e.g., McCoy, 2013). In particular, neurobiological models of adversity have posited that physiological arousal in response to threatening (e.g., violent) situations can lead children to enact more automatic, less reflective responses to external stimuli (McLaughlin & Sheridan, 2016; Ursache, Blair, & Raver, 2012). In turn, these disruptions in children's basic regulatory processing may have broader consequences for their everyday behavior and performance (e.g., ability to control aggressive impulses like those measured on the SDQ or to focus during a developmental assessment like the PRIDI). Although we lack the necessary longitudinal and physiological data to test these mediational hypotheses, future work exploring the mechanisms underlying impacts of community violence would be useful for identifying potential levers for intervention.

Importantly, even though the effects of community violence did vary across space and time, they remained statistically detectable even when the violent events took place as far as 1 km from children's homes or as long ago as 4 weeks. These findings stand in contrast to prior work, which has generally shown effect fade-out by 7–10 days following exposure (e.g., Sharkey, 2010; Sharkey et al., 2012). It is possible that the effects of community violence may have been particularly pervasive and long-lasting for this sample due to their age and setting. Indeed, prior theoretical models have emphasized the salience of both developmental age and cultural context when unpacking the effects of violence on children (Defoe, 2021; McCoy, 2013). Developmentally, children's brains are thought to be especially responsive to environmental threats during the first thousand days of life (Nelson, 2013), suggesting that exposure to community violence during the early years may be particularly consequential. Contextually, relative to their peers from higher-income contexts, children in our sample were likely exposed to fewer supports to offset their negative experiences of violence exposure. Similarly, the

caregivers of children in our sample may have had fewer resources to manage their own regulatory or behavioral responses to these acute stressors, limiting their ability to provide emotional support or to model positive coping strategies for their children (Sharkey et al., 2012).

Limitations

This study is not without limitations. First, regarding internal validity, despite making use of natural variation in community homicides, covariates (including chronic violence exposure) to account for potential imbalance, district fixed effects, and sensitivity analyses intended to maximize comparability between groups, it is possible that our samples of exposed and unexposed children differed based on unobserved factors that may have biased results. Accordingly, these findings should be interpreted with caution and future research should make additional attempts to mitigate selection issues (e.g., by controlling for children's development and behavior pre-exposure, which was not possible in this study). Second, regarding external validity, results cannot be generalized beyond the low-income population living in these seven districts in São Paulo. Although our study is one of the few to examine acute violence exposure outside of a high-income country, further research is needed to understand the applicability of these findings to other culturally and socioeconomically diverse populations around the world. Third, two of our three child outcomes were measured using parents' reports of child behavior in the preceding 6 months, which extends well before the 2–4 weeks during which violence exposure was measured. Prior work suggests that respondents' reports largely reflect recent behaviors (Walentynowicz, Schneider, & Stone, 2018). If parents were in fact considering children's longer-term behavior in their reports, we expect that this temporal mismatch would lead to an underestimation of the true effects of recent violence on children. Accordingly, our results should be replicated using narrow reporting windows that align with the definitions of exposure used. Fourth, our study considered specific definitions of community violence. Future research considering other forms of community violence, different definitions of exposure (e.g., chronic; direct vs. indirect), and additional child outcomes would be useful. Similarly, research exploring potential mechanisms underlying violence's effects on children (e.g., physiological reactivity, caregiver responsiveness, community/police reactions) is needed.

Conclusions

Overall, these findings suggest that community violence can have meaningful short-term impacts on young children's behavioral and developmental outcomes, with potentially more durable effects on

their regulatory skills. Although our analytic approach precludes exploration of the long-term implications of exposure, the specific mechanisms underlying these impacts, and the differential effects of varying forms of exposure, it does provide an overall estimate of the ways that both direct and indirect exposure to community violence may acutely impact children, including those who are not consciously aware that the violence has taken place. These results highlight the need for both indicated interventions for individuals experiencing the direct consequences of violent victimization, as well as more universal programming for families with young children living in violent areas.

Supporting information

Additional supporting information may be found online in the Supporting Information section at the end of the article:

Appendix S1. Details of child outcome measures.

Table S1. Baseline characteristics of children in the original ROC cohort and children included in the present analytic sample.

Table S2. Summary statistics and psychometric properties of the effortful control subscale of the Early Childhood Behavior Questionnaire (ECBQ).

Table S3. Summary statistics and psychometric properties of the Strengths & Difficulties Questionnaire (SDQ).

Table S4. Summary statistics and psychometric properties of the Engle scale of the Regional Project on Child Development Indicators (PRIDI).

Table S5. Distribution of children in the exposed and unexposed groups within 1 week before the child's three-year assessment.

Table S6. Distribution of children in the exposed and unexposed groups within 2 weeks before the child's three-year assessment.

Table S7. Distribution of children in the exposed and unexposed groups within 3 weeks before the child's three-year assessment.

Table S8. Distribution of children in the exposed and unexposed groups within 4 weeks before the child's three-year assessment.

Table S9. Sample descriptive statistics and balance across community violence-unexposed and exposed children (600 m and 3 weeks).

Table S10. Sample descriptive statistics and balance across community violence-unexposed and exposed children (600 m and 4 weeks).

Table S11. Sample descriptive statistics and balance across community violence-unexposed and exposed children (700 m and 2 weeks).

Table S12. Sample descriptive statistics and balance across community violence-unexposed and exposed children (700 m and 3 weeks).

Table S13. Sample descriptive statistics and balance across community violence-unexposed and exposed children (700 m and 4 weeks).

Table S14. Sample descriptive statistics and balance across community violence-unexposed and exposed children (800 m and 2 weeks).

Table S15. Sample descriptive statistics and balance across community violence-unexposed and exposed children (800 m and 3 weeks).

Table S16. Sample descriptive statistics and balance across community violence-unexposed and exposed children (800 m and 4 weeks).

Table S17. Sample descriptive statistics and balance across community violence-unexposed and exposed children (900 m and 2 weeks).

Table S18. Sample descriptive statistics and balance across community violence-unexposed and exposed children (900 m and 3 weeks).

Table S19. Sample descriptive statistics and balance across community violence-unexposed and exposed children (900 m and 4 weeks).

Table S20. Sample descriptive statistics and balance across community violence-unexposed and exposed children (1 km and 2 weeks).

Table S21. Sample descriptive statistics and balance across community violence-unexposed and exposed children (1 km and 3 weeks).

Table S22. Sample descriptive statistics and balance across community violence-unexposed and exposed children (1 km and 4 weeks).

Table S23. Effects on self-regulation (effortful control, as measured by the ECBQ).

Table S24. Effects on behavior (total behavior problems, as measured by the SDQ).

Table S25. Effects on child development (as measured by the PRIDI).

Table S26. Subgroup effects of community violence exposure on child outcomes, by child's sex.

Table S27. Subgroup effects of community violence exposure on child outcomes, by caregiver education.

Table S28. Subgroup effects of community violence exposure on child outcomes, by household SES.

Table S29. Subgroup effects of community violence exposure on child outcomes, by receipt of Bolsa Familia social assistance program.

Table S30. Subgroup effects of community violence exposure on child outcomes, by social support.

Table S31. Subgroup effects of community violence exposure on child outcomes, by chronic community violence exposure.

Table S32. Distribution of children in the exposed and unexposed groups within 2 weeks before and after the child's three-year assessment.

Table S33. Distribution of children in the exposed and unexposed groups within 3 weeks before and after the child's three-year assessment.

Table S34. Distribution of children in the exposed and unexposed groups within 4 weeks before and after the child's three-year assessment.

Figure S1. Estimates (with 90% confidence intervals) of the acute effect of exposure to recent community violence on various subdomains of children's outcomes across different geographic radii and temporal windows.

Figure S2. Estimates (with 90% confidence intervals) of the acute effect of exposure to recent community

violence on children's outcomes across different geographic radii and temporal windows, using listwise deletion for child outcomes.

Figure S3. Estimates (with 90% confidence intervals) of the acute effect of exposure to recent community violence on children's outcomes across different geographic radii and temporal windows, using a comparison group of unexposed children who did experience violence after assessment.

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Ethics and data availability

The original ROC study as well as all follow-up rounds received ethical approval from HUUSP's Ethics Committee for the Analysis of Research Projects (01604312.1.0000.0065). Informed consent was received from parents for their and their children's participation in the ROC study. Ethical approval for the present analyses was obtained from the Harvard University Area Institutional Review Board (IRB00000109). Requests for all data and materials used for this study can be sent via email to the last author at alexandra@usp.br.

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Key points

- On average, children who are exposed to adversity in their homes tend to demonstrate more negative behavioral and developmental outcomes than their peers who do not face such risks. However, relatively little is known about how adversity in the community might affect very young children, especially those outside of high-income countries.
- Our study examined the impacts of community violence on low-income 3-year-olds living in seven neighborhoods in São Paulo, Brazil.
- We found that children who were assessed soon after a homicide took place in their community showed lower self-regulation, higher behavior problems, and lower performance on a developmental assessment compared with children from the same neighborhoods who had not recently experienced community violence.
- These results suggest the need to support not only victims and witnesses of community violence but also children living in violence-affected areas more broadly.

References

- Anderson, M.L. (2008). Multiple inference and gender differences in the effects of early intervention: A reevaluation of the abecedarian, Perry preschool, and early training projects. *Journal of the American Statistical Association*, 103, 1481–1495.
- Brentani, A., Scoleze Ferrer, A.P., Brentani, H., Liu, C.H., Grisi, S.J., Valente, M.H., ... & Fink, G. (2020). Cohort profile: São Paulo Western region birth cohort (ROC). *International Journal of Epidemiology*, 49, 1438.
- Britto, P.R., Lye, S.J., Proulx, K., Yousafzai, A.K., Matthews, S.G., Vaivada, T., ... & Lancet Early Childhood Development Series Steering Committee. (2017). Nurturing care: Promoting early childhood development. *The Lancet*, 389, 91–102.
- Cardena, M.M., Ribeiro-dos-Santos, A., Santos, S., Mansur, A.J., Pereira, A.C., & Fridman, C. (2013). Assessment of the relationship between self-declared ethnicity, mitochondrial haplogroups and genomic ancestry in Brazilian individuals. *PLoS One*, 8, e62005.
- Chen, P., Voisin, D.R., & Jacobson, K.C. (2016). Community violence exposure and adolescent delinquency: Examining a spectrum of promotive factors. *Youth & Society*, 48, 33–57.
- Cuartas, J., & Leventhal, T. (2020). Exposure to community violence and Children's mental health: A quasi-experimental examination. *Social Science & Medicine*, 246, 112740.
- Cuartas, J., McCoy, D.C., & Molano, A. (2018). The acute effect of community violent crime on maternal engagement in cognitive and socioemotional stimulation. *Early Childhood Research Quarterly*, 45, 143–154.
- Defoe, I.N. (2021). Towards a hybrid criminological and psychological model of risk behavior: The developmental neuro-ecological risk-taking model (DNERM). *Developmental Review*, 62, 100995.
- Fernald, L.C., Kariger, P., Hidrobo, M., & Gertler, P.J. (2012). Socioeconomic gradients in child development in very young children: Evidence from India, Indonesia, Peru, and Senegal. *Proceedings of the National Academy of Sciences of the United States of America*, 109, 17273–17280.
- Filmer, D., & Pritchett, L.H. (2001). Estimating wealth effects without expenditure data—or tears: An application to educational enrollments in states of India. *Demography*, 38, 115–132.
- Finkelhor, D., Turner, H.A., Shattuck, A., & Hamby, S.L. (2015). Prevalence of childhood exposure to violence, crime, and abuse: Results from the national survey of children's exposure to violence. *JAMA Pediatrics*, 169, 746–754.
- Frongillo, E.A., Kulkarni, S., Basnet, S., & de Castro, F. (2017). Family care behaviors and early childhood development in low-and middle-income countries. *Journal of Child and Family Studies*, 26, 3036–3044.

- Goodman, R. (1997). The strengths and difficulties questionnaire: A research note. *Journal of Child Psychology and Psychiatry*, 38, 581–586.
- Hamadani, J.D., Tofail, F., Hilaly, A., Huda, S.N., Engle, P., & Grantham-McGregor, S.M. (2010). Use of family care indicators and their relationship with child development in Bangladesh. *Journal of Health, Population, and Nutrition*, 28, 23.
- Hillis, S., Mercy, J., Amobi, A., & Kress, H. (2016). Global prevalence of past-year violence against children: A systematic review and minimum estimates. *Pediatrics*, 137, e20154079.
- Instituto Brasileiro de Geografia e Estatística (IBGE). (2010). 2010 Census. Available from: <https://www.ibge.gov.br/en/statistics/social/education/18391-2010-population-census.html?edicao=19327&t=resultados> [last accessed 1 June 2022].
- Kahle, D., & Wickham, H. (2013). Ggmap spatial visualization with ggplot2. *The R Journal*, 5, 144–161.
- Kariger, P., Frongillo, E.A., Engle, P., Britto, P.M.R., Sywulka, S.M., & Menon, P. (2012). Indicators of family care for development for use in multicountry surveys. *Journal of Health, Population and Nutrition*, 30, 472–486.
- Lall, R. (2016). How multiple imputation makes a difference. *Political Analysis*, 24, 414–433.
- Leventhal, T. (2018). Neighborhood context and children's development: When do neighborhoods matter most? *Child Development Perspectives*, 12, 258–263.
- Leventhal, T., Dupéré, V., & Shuey, E.A. (2015). Children in neighborhoods. In R.M. Lerner (Ed.), *Handbook of child psychology and developmental science: Vol. 4. Ecological settings and processes* (7th edn, pp. 493–533). Hoboken, NJ: Wiley.
- McCleary, L. (2002). Using multiple imputation for analysis of incomplete data in clinical research. *Nursing Research*, 51, 339–343.
- McCoy, D.C. (2013). Early violence exposure and self-regulatory development: A bioecological systems perspective. *Human Development*, 56, 254–273.
- McCoy, D.C., Peet, E.D., Ezzati, M., Danaei, G., Black, M.M., Sudfeld, C.R., ... & Fink, G. (2016). Early childhood developmental status in low-and middle-income countries: National, regional, and global prevalence estimates using predictive modeling. *PLoS Medicine*, 13, e1002034.
- McCoy, D.C., Raver, C.C., & Sharkey, P. (2015). Children's cognitive performance and selective attention following recent community violence. *Journal of Health and Social Behavior*, 56, 19–36.
- McLaughlin, K.A., & Sheridan, M.A. (2016). Beyond cumulative risk: A dimensional approach to childhood adversity. *Current Directions in Psychological Science*, 25, 239–245.
- McLaughlin, K.A., Weissman, D., & Bitrán, D. (2019). Childhood adversity and neural development: A systematic review. *Annual Review of Developmental Psychology*, 1, 277–312.
- Mohammad, E.T., Shapiro, E.R., Wainwright, L.D., & Carter, A.S. (2015). Impacts of family and community violence exposure on child coping and mental health. *Journal of Abnormal Child Psychology*, 43, 203–215.
- Molano, A., Harker, A., & Cristancho, J.C. (2018). Effects of indirect exposure to homicide events on children's mental health: Evidence from urban settings in Colombia. *Journal of Youth and Adolescence*, 47, 2060–2072.
- Nelson, C.A. (2013). Biological embedding of early life adversity. *JAMA Pediatrics*, 167, 1098–1100.
- Obando, D., Wright, N., & Hill, J. (2023). Warmth and reciprocity with mothers, and young children's resilience to exposure to community violence in Colombia: Findings from the La Sabana Parent-Child Study. *Journal of Child Psychology and Psychiatry*, 64, 197–205. <https://doi.org/10.1111/jcpp.13629>
- Putnam, S.P., Gartstein, M.A., & Rothbart, M.K. (2006). Measurement of fine-grained aspects of toddler temperament: The Early Childhood Behavior Questionnaire. *Infant Behavior and Development*, 29, 386–401.
- QGIS Development Team. (2021). QGIS geographic information system. Open Source Geospatial Foundation Project. Available from: <http://qgis.osgeo.org>
- Sharkey, P. (2010). The acute effect of local homicides on children's cognitive performance. *Proceedings of the National Academy of Sciences of the United States of America*, 107, 11733–11738.
- Sharkey, P.T., Tirado-Strayer, N., Papachristos, A.V., & Raver, C.C. (2012). The effect of local violence on children's attention and impulse control. *American Journal of Public Health*, 102, 2287–2293.
- Shonkoff, J.P., Boyce, W.T., & McEwen, B.S. (2009). Neuroscience, molecular biology, and the childhood roots of health disparities: Building a new framework for health promotion and disease prevention. *JAMA*, 301, 2252–2259.
- StataCorp. (2021). *Stata statistical software: Release 17*. College Station, TX: StataCorp.
- UN Office on Drugs & Crime. (2021). *Intentional homicides (per 100,000 people)*. Washington, DC: The World Bank. Available from: <https://data.worldbank.org/indicator/VC.IHR.PSRC.P5>
- Ursache, A., Blair, C., & Raver, C.C. (2012). The promotion of self-regulation as a means of enhancing school readiness and early achievement in children at risk for school failure. *Child Development Perspectives*, 6, 122–128.
- Verdisco, A., Cueto, S., Thompson, J., Engle, P., Neuschmidt, O., Meyer, S., ... & Miranda, A. (2009). *Urgency and possibility results of PRIDI: A first initiative to create regionally comparative data on child development in four Latin American countries technical annex*. Washington, DC: Technical Annex Inter-American Development Bank.
- Walentynowicz, M., Schneider, S., & Stone, A.A. (2018). The effects of time frames on self-report. *PLoS One*, 13, e0201655.

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